

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA15
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COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 50

Code of Conduct

I H MOHAMED FAAIZ 192524099 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1. Compare type-1 and type-2 Hypervisors

Performance:-

Ans: A type 1 hypervisor runs directly on the physical hardware while a type-2 hypervisor runs on top of a host operating system

Security :-

Type-1 offers stronger security with better VM isolation and a smaller attack surface .

Type-2 is easier to install but is mainly used for testing development and personal use

Manageability:-

Type-1 supports centralized management like migration, load balancing and high availability

Type:-2 is easier to install but is mainly used for testing development and personal use

Conclusion:-

Type 1 hypervisors are the better choice for mission critical cloud datacenters because they provide high performance, better security, improved scalability and reliable management

2. Virtualization Architecture for Workload Isolation

Virtualization allows multiple virtual machines (VMs) to run on the same physical server while keeping them isolated.

Each VM has its own operating system, memory, storage, and virtual network, preventing one workload from accessing another.

This architecture enables finance, healthcare, and student portal applications to securely share the same hardware, virtual switches, VLANs, and storage. Virtualizations further improve isolation and protect sensitive data.

There are two major attack surfaces.

Two major attack services are

1. Hypervisor attacks:-

Attackers exploit vulnerabilities in the hypervisor to gain control of multiple VMs

2. VM ESCAPE:

A malicious VM breaks out of its isolated environment and attacks the host or neighboring VMs

Conclusion: Strong isolation regular updates encryption and access control ensure multi tenant cloud environments

3. VM Host Utilization Analysis

The VM host shows CPU usage of 92%, memory usage of 88%, storage I/O latency of 35 ms and an average VM boot time of 170 seconds. These values indicate that the host is under heavy load.

High CPU usage slows application processing while high memory usage may lead to swapping and reduced performance. High storage latency indicates that disk operations are slow which increases VM boot time.

Corrective Actions:

Add more CPU and RAM
storage for SSD/NVMe drives distribute
workloads across additional hosts and
remove unnecessary or idle virtual machines

Continuous monitoring should be used to prevent resource bottlenecks

4. Software Defined Data center (SDDC)

Software Defined Data Center (SDDC)
Virtualizes compute, storage and Networking, allowing resources to be managed through Software instead of physical hardware. This makes the datacenter more flexible than a traditional data center

Compute virtualizations allows quick VM creation and migration storage Virtualization combine multiple storage devices into a single pool for a better Utilization

Network virtualization creates virtual network that can be configured without changing physical cables or switches

Conclusion : SDN improve agility scalability automation and resource utilization while reducing deployment time and operation costs compared to traditional data centers

5) Multi Cloud Federation identify Mis match

Identify mismatches occurs when different cloud providers use different authentication systems and are identified. This can prevent users from accessing services across multiple cloud platforms.

Secure solution is to implement identify using single Sign ON (SSO) with standards such as SAML, OAuth, 2.0, or OpenID connect. Security can be strengthened with Multifactor Authentication (MFA), Role Based Access Control (RBAC) and encryption to protect user information.

Conclusion:- Identity federation provides secure authentication across multiple cloud providers improves user experience with single login access and protects user privacy and sensitive data.